

# Practice Quiz 06B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 06

*A unique set with a property quantified over all subsets.*

Label and prove both halves.

1. Let  $U$  be a set. Prove that there is exactly one subset  $E$  of  $U$  such that  $X \cap E = X$  for every subset  $X$  of  $U$ .

**Solution** *Existence.* Take  $E = U$ . This is a subset of  $U$ . For every  $X \subseteq U$ , we have  $X \cap U = X$ , so  $E$  has the required property.

*Uniqueness.* Suppose  $E \subseteq U$  has the stated property. Taking  $X = U$  gives  $U \cap E = U$ . Since  $E \subseteq U$ , we also have  $U \cap E = E$ . Therefore  $E = U$ . Every subset with the required property equals the one constructed above, so it is unique. ■